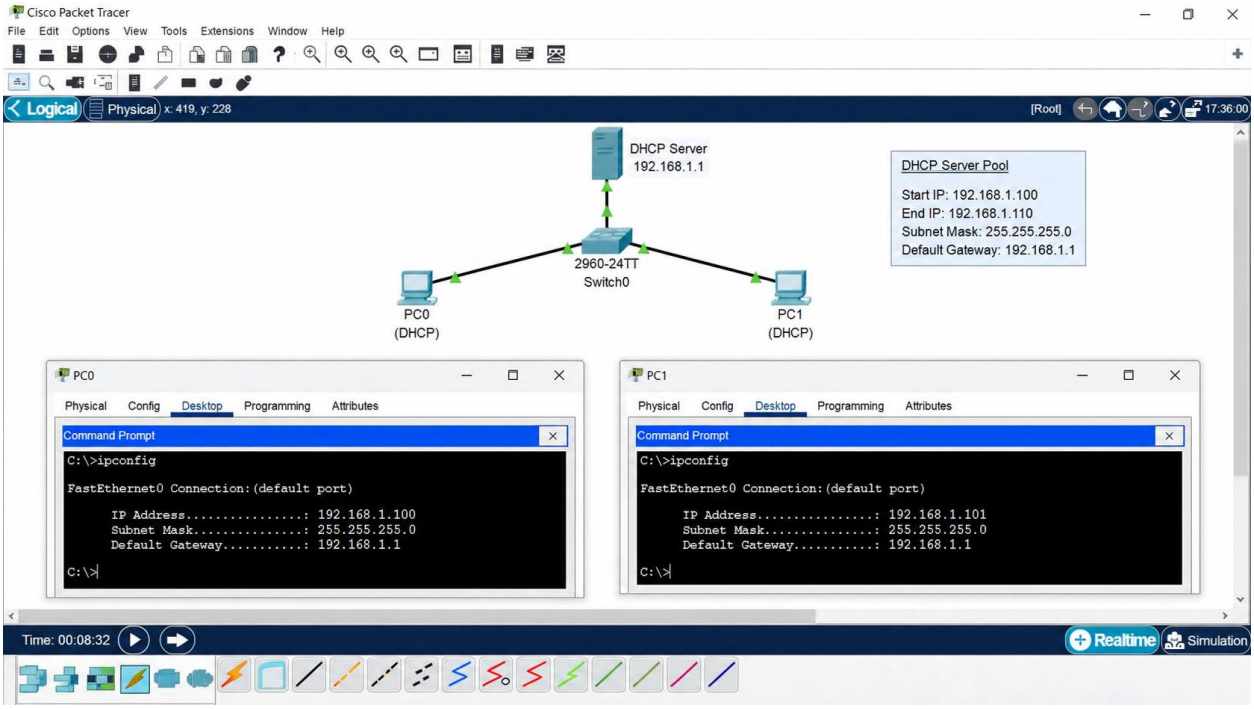
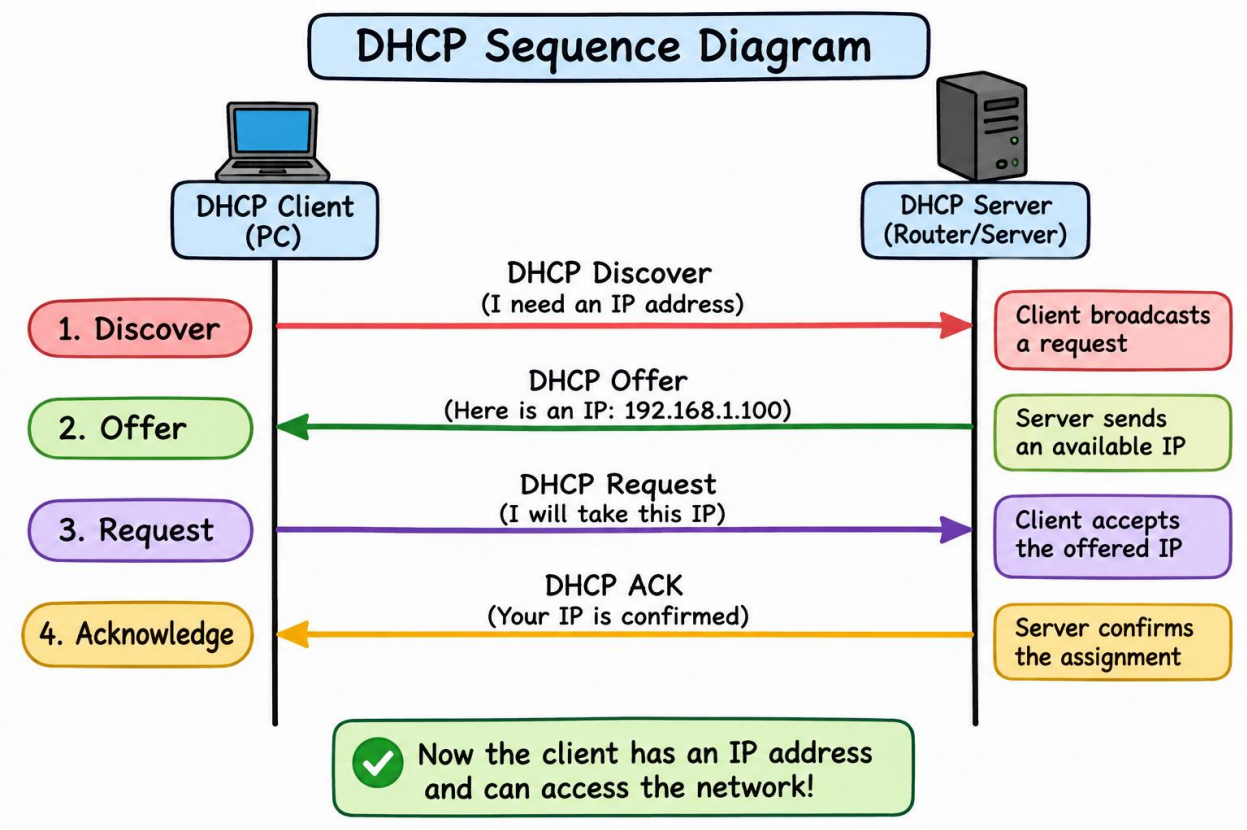
1)In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

Ans.



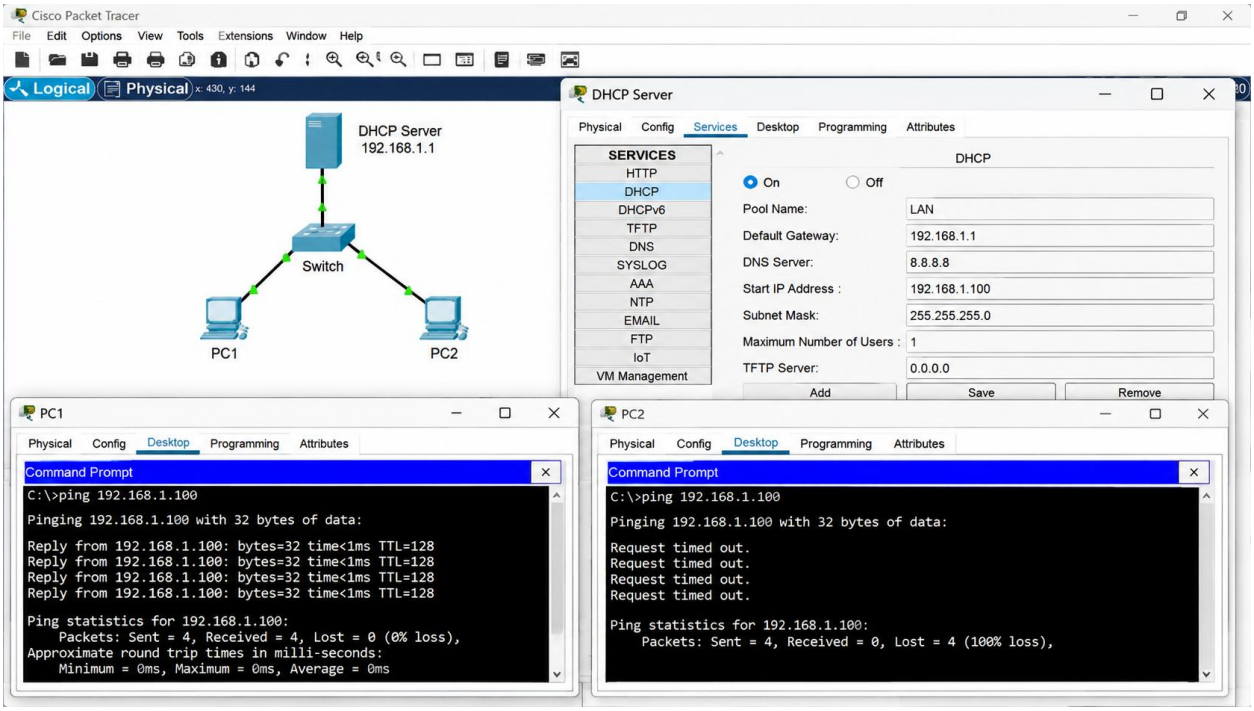
2)Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage Use arrows to show the direction of each message between client and server

Ans.



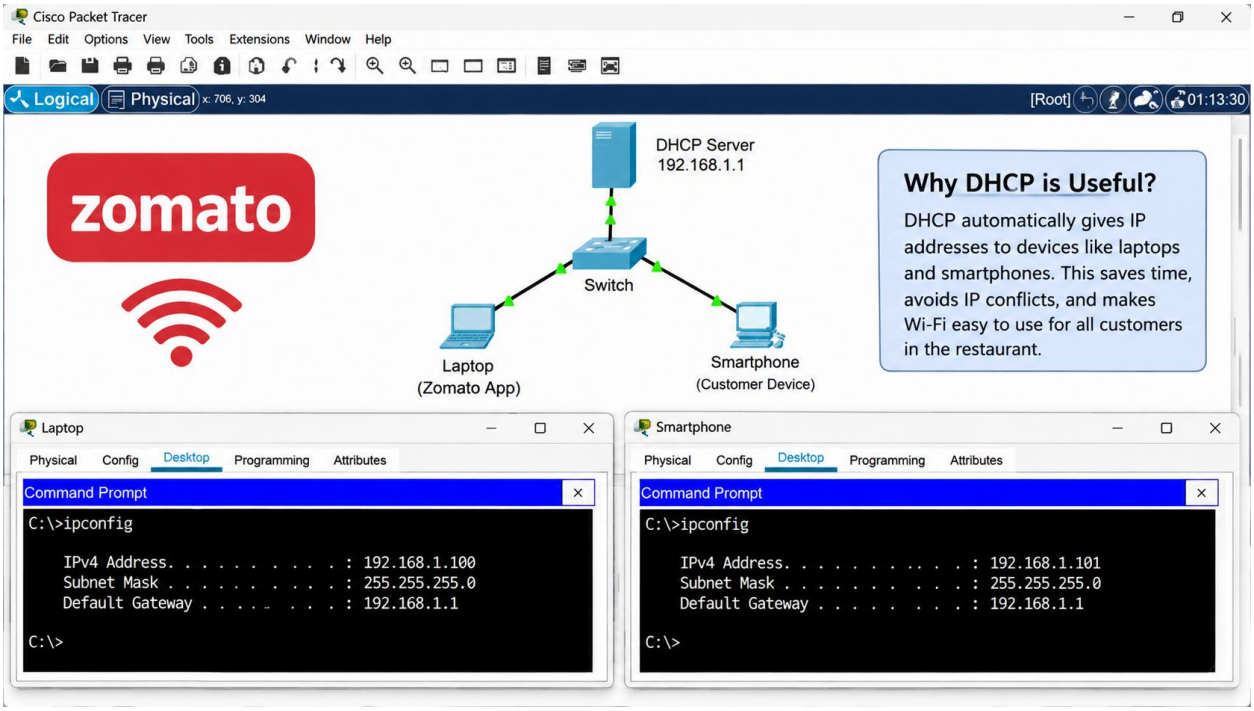
3)Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

Ans.



4)Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

Ans.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical Physical x: 706, y: 304
[Root] 01:13:30
zomato
DHCP Server
192.168.1.1
Switch
Laptop
(Zomato App)
Smartphone
(Customer Device)
Why DHCP is Useful?
DHCP automatically gives IP addresses to devices like laptops and smartphones. This saves time, avoids IP conflicts, and makes Wi-Fi easy to use for all customers in the restaurant.
Laptop
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig
IPv4 Address. . . . . : 192.168.1.100
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
C:\>
Smartphone
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ipconfig
IPv4 Address. . . . . : 192.168.1.101
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
C:\>